

Data Structures and Algorithms - Assignment 8

1. Implement heap sort using min heap.
2. <https://leetcode.com/problems/kth-largest-element-in-an-array/description> (solve using heap)
 - You will need 0 based indexing for heap
 - root is at `arr[0]`
 - node : `i`
 - left child : $2i + 1$
 - right child : $2i + 2$
 - parent : $(i-1)/2$
 - Last Non leaf node : $(n/2)-1 \Rightarrow$ first parent from right side
3. Implement BFS and DFS traversal for graph implemented using adjacency list method.